

6 (a) (i) The spread of the electron beam to form bright rings due to diffraction, B1

The presence of dark regions between bright rings indicate that destructive interference is taking place B1

(ii) 1. $KE = eV = p^2/2m$ M1

$$\lambda = \frac{h}{p}$$

$$= \frac{h}{\sqrt{2meV}}$$

M1

A0

2. $\lambda = \frac{h}{\sqrt{2meV}}$

$$= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(250)}}$$

M1

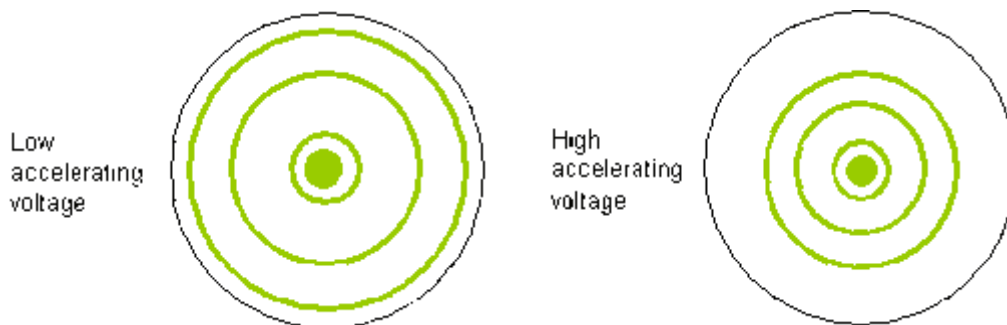
$$= 7.77 \times 10^{-12} \text{ m}$$

A1

(iii) As velocity increases, the de Broglie wavelength decreases since $\lambda \propto \frac{1}{\sqrt{V}}$. B1

B1

Since $d \sin \theta = n\lambda$, the angles at which the maxima are produced decrease. The rings become smaller/closer together.



(b) $\Delta p \Delta x \gtrsim h$

$$\Delta p (1.2) \gtrsim 6.63 \times 10^{-34}$$

$$\Delta p \gtrsim 5.53 \times 10^{-34} \text{ kg m s}^{-1}$$

M1

$$p = 80 \times 2.0 = 160 \text{ kg m s}^{-1}$$

Hence, the uncertainty in his momentum is negligible as compared to his linear momentum. B1

OR

[Turn over]

6

$$\tan \theta = \frac{\Delta p}{p} = \frac{5.53 \times 10^{-34}}{160} = 3.46 \times 10^{-36} \approx 0$$

Hence, the angle of deflection is negligible.